LUKAS PETERMANN

Heilbronn, DE
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EDUCATION

M.Sc. Computer Science, Otto-von-Guericke University, Magdeburg — Oct. 2024 - Sep. 2026
Master Thesis: *Investigating Matrix Assembly for High Performance CFD Codes, 1.0*

M.Sc. Computer Science, Karlsruhe Institute of Technology, Karlsruhe — Apr. 2023 - Sep. 2024

B.Sc. Computer Science, Otto-von-Guericke University, Magdeburg — Oct. 2018 - Mar. 2023
Bachelor Thesis: *Comparison of Real-Time Plane Detection Algorithms on Intel RealSense, 1.0*

A Levels, Herderschule Rendsburg — 2008 - 2017

SKILLS

Technical Skills Julia, Kotlin, Python, C++, Git, Flask, NumPy, Open3D, Linux, Spring Boot, Jetpack Compose, Heimdall, OpenFGA, Apache Kafka

EXPERIENCES

PhD Student — Oct. 2026 - Today
Technical University of Munich, Campus Heilbronn — *Heilbronn, DE*

- Researching High-performance CFD and Julia.

Working Student Full-Stack Software Development — Apr. 2024 - Aug. 2026
SoftwareBees GmbH — *Krebeck, DE*

- Misavo: The digital twin for hospitality and event organization.
  - Developed full-stack features across authentication, multi-tenancy, inventory, ordering, and workflow automation.
  - Worked across UI, APIs, authentication, and quality assurance.
- Vicus: The mobile app to digitalize living in communities.
  - Developed the front-end using Jetpack Compose
  - Designed the middle-end using a combination of Heimdall and OpenFGA.
  - Implemented the backend using Ktor in a hexagonal architecture.
- Integrated the text extraction (see *Freelancing Work*) into a kafka pipeline and added a prior document splitting step.
- Developed an automated service that processes supplier data from SAP-based Excel files and sends itemized emails using Kotlin, Apache POI, and Spring Batch.

Research Assistant — Apr. 2023 - Sep. 2023
Karlsruhe Institute of Technology, Steinbuch Centre for Computing — *Karlsruhe, DE*

- Developed and maintained the *OpenFOAM Benchmark Runner* in Python.

Summer Intern — Oct. 2023 - Nov. 2023
The Mathworks GmbH, Parallel Computing Team — *Munich, DE*

- Implemented sparse triangular preconditioners for GPU iterative solvers.
- Improved the performance of GPU sparse `mldivide` using cuSPARSE.

Research Assistant Jan. 2021 - Mar. 2024
Otto-von-Guericke University, Department of Software Engineering *Magdeburg, DE*

- Implemented a time- and memory-efficient version of RSPD in Python.
- Designed a simulator for SLAM algorithms in Python.
- Implemented a 3D-Minimap within the context of an AR Unity app in C#.

Intern Aug. 2020 - Dec. 2020
LIVEDAB GmbH *Hamburg, DE*

- Developed new features within a filter marketing platform using Typescript and React.
- Designed a RESTful API and database model using PostgreSQL.

Working Student Software Development Oct. 2019 - Dec. 2019
BM Sports Technology GmbH *Magdeburg, DE*

- Extended the User-Interface of an iOS app using SwiftUI.
- Implemented real-time visualization of sensor data using SwiftUI.

FREELANCING WORK

SoftwareBees GmbH Feb. 2024 - Mar. 2024

- Developed automatic extraction of textual data using Python, Tesseract, and OpenCV with a 90% success rate.

XCO-FDK Mar. 2023 - Jun. 2023

- Developed automatic Code Generation using Python and Jinja2.
- Set up semi-automatic continuous documentation using Python and Sphinx.

JD-Software Aug. 2023

- Co-created a Spring-Boot development guide.

SOFTWARE

JuNe.jl Julia matrix assembly embedded into NeoN as part of my master's thesis.

RSPyD Python implementation of RSPD with focus on efficiency.

CYK Validator Application that transforms a context-free grammar into Chomsky-Normalform and reports all substrings of a given word that can be generated by said grammar.

OBRG-Py A Python implementation of the *Octree-based Region Growing* plane detection algorithm as part of my bachelor thesis.

Torte-Dashboard A dashboard for the visualization of KConfig-based project metrics and plots. The pre-processing of *Torte* experiment data is written in Python, while the frontend is built using vuetify and Typescript.

CERTIFICATES

Node Level Performance Engineering June 2023
High-Performance Computing Centre Stuttgart *Stuttgart, DE*

PUBLICATIONS

Testing Strategies for OpenFOAM Projects Gärtner, Jan & Olenik, Gregor & Fadeli, Mohammed & Petermann, Lukas & Kronenburg, Andreas & Marschall, Holger & Anzt, Hartwig. (2025). OpenFOAM® Journal. 5. 115-130. 10.51560/ofj.v5.134.

EXTRA-CURRICULAR ACTIVITIES

Presented preliminary master's findings at OpenFOAM Workshop July 2026
Guimaraes, POR

High-Performance Computing Status Conference Attendee June 2023
Karlsruhe Institute of Technology *Karlsruhe, DE*

Student council member of the Faculty of Computer Science Sep. 2020 - Mar. 2021
Otto-von-Guericke University Magdeburg *Magdeburg, DE*

LEADERSHIP

Organized the mentoring program at the Faculty of Computer Science. Sep. 2020 - Mar. 2021
Otto-von-Guericke University Magdeburg *Magdeburg, DE*